

Che Cui

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INTRODUCTION

I hold a BSc in Data Science and Artificial Intelligence from the Technical University of Madrid. I have developed a strong foundation in programming, mathematics, and statistics, complemented by hands-on experience in both the private and public sectors. To deepen this focus, I decided to pursue an MRes in AI and Machine Learning in Imperial College London, seeking a new environment. I am motivated to continue learning and contributing to the field of artificial intelligence. My research interests lie at the intersection of machine learning, computer vision, and natural language processing.

EDUCATION

Imperial College London September 2025 – Present
MRes AI and Machine Learning London, UK

- Modules: Generative AI, Research Tutorial, Simulated Research and Development Project Proposal, Ethics, Fairness, and Explanation in AI among others.
- Individual Research Project: “Unsupervised anomaly detection for cardiac MRI”. The thesis is supervised by Dr. Giacomo Tarroni and co-supervised by Dr. Chen Qin. It is assessed by a poster presentation, a thesis and a subsequent oral presentation of the written thesis.

Technical University of Madrid (UPM) September 2020 - March 2025
Bachelor of Data Science and Artificial Intelligence Madrid, ES

- Modules: Deep Learning, NLP, Generative AI, Calculus, Linear Algebra, Statistics between others.
- Bachelor thesis: “Diffusion Model Based Brain Tumor Segmentation Enhanced with Inpainting Method” Grade: 10/10. This thesis was supervised by Dr. Guillermo Iglesias.
- The program’s duration was extended to the 2024-2025 academic year to allow for the completion of my thesis at home university following an exchange program at TUM.

Technical University of Munich (TUM) April 2024 - September 2024
Exchange Study Programme Munich, GER

- Modules: Advanced Natural Language Processing, Foundations and Applications of Graph Neural Networks.
- Lab project in Praktikum - Reinforcement Learning and Benchmarking for Modular Robots.

WORK EXPERIENCE

Research Assistant March 2025 – September 2025
Ontology Engineering Group at Technical University of Madrid Madrid, ES

- Research assistant to Dr. Elvira Amador Domínguez at the Ontology Engineering Group.
- Project: “GAP-KGE: Automatic Template Generation for Reasoning Models over Knowledge Graphs”

Data scientist intern October 2023 – January 2024
Renta 4 investment banking Madrid, ES

- Worked in the Digital Innovation department to develop and implement Machine Learning solutions.
- Projects: Investment Fund Recommender System & Automated sentiment Analysis and Report Generation Using NLP methods.

RESEARCH PROJECTS

Unsupervised anomaly detection for cardiac MRI | Imperial College London 2025 - 2026

- Supervised by Dr. Giacomo Tarroni and co-supervised by Dr. Chen Qin.
- Literature review on state-of-the-art unsupervised anomaly detection methods and test generative and self-supervised approaches on public datasets, enabling detection of structural anomalies (e.g. hypertrophic cardiomyopathy) and motion abnormalities without labelled pathology.
- Evaluation, poster preparation and final thesis drafting follow, culminating in thesis submission by end-August 2026.

- Worked closely with the Dr. Elvira Amador Domínguez on a project for information extraction from scientific papers using ML and LLM-based models to generate templates for Knowledge Graph Embedding papers.
- Literature review on state-of-the-art information extraction methods and set up of a GitHub repository and pipeline.

Diffusion Model Based Brain Tumor Segmentation Enhanced with Inpainting Method | *BSc thesis* 2025

- Worked in U-Net based diffusion models, also applied RePaint inpainting method, for image generation but focused on the image segmentation task for brain tumors using 2020 Brats Dataset. Official university link to the project [here](#).
- Literature review on state-of-the-art papers in image generation and inpainting methods for diffusion models.
- Set up of a GitHub repository and pipeline.
- Submitted for paper publication in Computer Methods and Programs in Biomedicine journal. Submission confirmation [here](#).

OTHER RELEVANT PROJECTS**CONSENSE** | *Practical Lab Project at TUM*

2024

- Project full name is “CONSENSE: Optimal Path Planning and Collision Avoidance for Robots using Sensor Fusion”. Collaborated in a 3 people team to develop a hardware and software selection for a industrial robot to work in concrete removal task. Developed a simulation environment in ROS2 and Gazebo to model and test the robot’s functionality in support of a Start Up project from TUM.
- Literature review on sensor fusion and validated sensor performance in harsh working conditions.
- Research report writing and presented the project findings in a poster session at a open TUM event in front of supervisors and PhD students.

COMPETITIONS

Competitive programming: 2nd place in the Nanyang Technological University Singapore - Imperial College Learning Analytics Hackathon 2025 at Imperial College London out of more than 20 teams. Proposed a solution on how to use real-world student learning data to support Imperial student’s learning.

Bachelor thesis: Best Thesis with Impact 2024-2025 award, given to the best thesis across all Master and Bachelor thesis across the School of Computer Systems Engineering.

Other: Winner of multiple math contests from primary to high school educations.

TECHNICAL SKILLS

Programming Languages: Python, C, R, Matlab, SQL & No-SQL (MongoDB, Neo4j, Cassandra).

Libraries: Keras, TensorFlow, Pytorch, NumPy, SciPy, Pandas, Matplotlib.

Relevant Programming Skills: LaTeX, Spark, Docker, GitHub, AWS certificate.

Languages: Spanish (Native), Chinese (Native), English (Cambridge C2 Proficiency), French (Basic proficiency).

INTEREST

Societies: Active member of the Department of Computing Society attending industry talks and regular member of the Cross Country & Athletics.

Volunteering: Acted as Degree Vice-Representative for the students in the Data Science and Artificial Intelligence bachelor program at Technical University of Madrid.

References available on request.